

Tests for equivalence or non-inferiority for paired binary data

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SUMMARY

Assessment of therapeutic equivalence or non-inferiority between two medical diagnostic procedures often involves comparisons of the response rates between paired binary endpoints. The commonly used and accepted approach to assessing equivalence is by comparing the asymptotic confidence interval on the difference of two response rates with some clinical meaningful equivalence limits. This paper investigates two asymptotic test statistics, a Wald-type (sample-based) test statistic and a restricted maximum likelihood estimation (RMLE-based) test statistic, to assess equivalence or non-inferiority based on paired binary endpoints. The sample size and power functions of the two tests are derived. The actual type I error and power of the two tests are computed by enumerating the exact probabilities in the rejection region. The results show that the RMLE-based test controls type I error better than the sample-based test. To establish an equivalence between two treatments with a symmetric equivalence limit of 0.15, a minimal sample size of 120 is needed. The RMLE-based test without the continuity correction performs well at the boundary point 0. A numerical example illustrates the proposed procedures. Copyright © 2002 John Wiley & Sons, Ltd.

KEY WORDS: asymptotic test; correlated binary endpoints; confidence interval; diagnostic tests; two one-sided tests

1. INTRODUCTION

Assessment for equivalence can generally be divided into two major objectives: an equivalence test and a non-inferiority test [1]. Generic drug clinical trials often involve test and reference comparisons with an objective to demonstrate the equivalence, within a tolerable interval, of the two formulations. On the other hand, it is often of interest to assess if a less toxic, easier to administer and/or less expensive test treatment is not inferior, by an acceptable small amount, to the reference treatment. Bioequivalence studies are typically designed as cross-over experiments with 30 or 40 patients in which the generic formulation and the innovative formulation of the same drug are compared with regard to the rate and extent of absorption. For some drug products that do not produce any appreciable absorption into

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the systemic circulation, the equivalence between the two formulations is directly assessed through therapeutic endpoints [2, 3].

An evaluation of the performance of a diagnostic procedure or screening test focuses on sensitivity and specificity. Sensitivity and/or specificity of the alternative (or reference) procedure can be evaluated against the gold standard. A gold standard procedure is assumed without classification error. Diagnostic procedures with both high sensitivity and high specificity are obviously desirable. The reference procedures are frequently too expensive, hazardous, and/or difficult to operate; a less sophisticated, cheaper or safer test procedure but with higher error rates, therefore, may be preferred if the loss of the sensitivity or specificity is within some acceptable limits.

Testing for equivalence or non-inferiority between two diagnostic methods often involves comparisons of the response rates of paired binary endpoints. For example, in comparing diagnostic efficacy of an non-invasive alternative diagnostic (test) procedure to an invasive (reference) method, the patients with suspected liver lesions first receive the alternative non-invasive procedure such as post-contrast ultrasound, post-contrast MRI, followed by the invasive method such as computer tomography arterio portography (CTAP). Sometimes, at the end of the study a histopathologic examination from biopsy is performed on each patient. The histopathological results from the biopsy will be served as the 'gold standard' for the diagnostic efficacy of both procedures. For this type of data, two binary clinical endpoints are observed on each patient. The two endpoints are correlated since they all are observed on the same patient.

Statistical procedures developed for testing equivalence or non-inferiority so far concentrate on comparing the rates of two independent binary responses [4–10]. Procedures for assessing equivalence or inferiority between the paired binary endpoints have not been well studied. McNemar's test [11] has been used to detect a difference between paired binary endpoints. The power and sample size formulae for McNemar's test have been extensively examined by many investigators [12–15]. However, these methods are not applicable when the objective is to establish equivalence or non-inferiority. Recently Lu and Bean [16] and Nam [17] proposed test statistics and sample size determination for comparing two diagnostic methods in the context of the (one-sided) non-inferiority test of sensitivity. A non-inferiority test imposes an implicit assumption that the reference procedure is a 'gold standard' procedure. In other words, in the presence of a gold standard procedure, we are interested in whether the test procedure is almost as effective as (effectively not inferior to) the gold (reference) standard procedure.

Recently, Lachenbruch and Lynch [18] considered issues in assessing an equivalence between two test diagnostic procedures without a gold standard procedure. Assessing equivalence between two test methods without access to a gold standard arises in many situations. For example, an investigator wishes to examine whether a particular body fluid gives results equivalent to the results of using plasma in an HIV screening test (there was no gold standard available). In the absence of a gold standard, assessment of equivalence considers a hypothesis that the alternative procedure gave either more or fewer positive (negative) results than the reference procedure. Lachenbruch and Lynch [18] proposed a likelihood ratio test for equivalence in terms of the rate ratio of the two procedures. In this paper, we extend the Lu and Bean [16] sample statistic and the Nam [17] score statistic for a (one-sided) non-inferiority test to a (two-sided) equivalence test in terms of the rate difference.

Given a specific equivalence limit, δ , the assessment of equivalence between an alternative procedure and a reference procedure is formulated by the null hypothesis that the difference

between the success rates of the two treatments is outside the interval $(-\delta, \delta)$ against the alternative hypothesis that the difference is within this interval. The test can be performed by simultaneously testing two sets of one-sided hypothesis each at the nominal level α . To conclude the equivalence, both hypotheses need to be rejected. Consequently, the two one-sided tests form a test with an overall type I error rate α . Alternatively, a bioequivalence study is often tested by comparing the $100(1 - 2\alpha)$ per cent confidence limits of the difference with the equivalence limits $(-\delta, \delta)$ [1]. Equivalence is inferred if the entire confidence interval falls within the equivalence limits. For the non-inferiority test, the one-sided null hypothesis is that the difference is less than or equal to $-\delta$. The corresponding one-sided confidence interval can be derived accordingly. The confidence interval approach is recommended by regulatory agencies.

In this paper, we investigate both the confidence interval and hypothesis testing approaches to assessing equivalence or non-inferiority between paired binary endpoints. In Section 2, a Wald-type sample-based test statistic [16] and the RMLE-based (score) test statistic [17] are given for the hypothesis of therapeutic equivalence in terms of two one-sided tests. In Section 3, we compare the two tests with respect to the exact Type I error and powers for various conditions. In addition, we derive the power function and sample size for the two test statistics. In Section 4, numerical examples illustrate the proposed methods, with and without a gold standard, for comparing the diagnostic efficacy of the two diagnostic procedures. Discussion and final remarks are provided in Section 5.

2. TEST STATISTICS

2.1. Interval Hypotheses

Let X_{0j} and X_{1j} , $j=0,1$, be the binary responses for the alternative and reference invasive procedures, respectively. Denote the total number of patients as n . Let x_{00} , x_{01} , x_{10} and x_{11} be the observed numbers of pairs (0,0), (0,1), (1,0) and (1,1), respectively; x_{10} and x_{01} are the discordant pairs, and x_{00} and x_{11} are the concordant pairs. The four outcomes and probabilities can be classified into a 2×2 table:

Diagnosis for test procedure	Diagnosis for reference procedure		Total
	Yes (1)	No (0)	
Yes (1)	x_{11} (p_{11})	x_{10} (p_{10})	x_t (p_t)
No (0)	x_{01} (p_{01})	x_{00} (p_{00})	$n - x_t$ ($1 - p_t$)
Total	x_r (p_r)	$n - x_r$ ($1 - p_r$)	n

Note that the joint distribution of x_{10} and x_{01} is a trinomial with response rates p_{10} and p_{01} , respectively.

The 2×2 table represents a typical schema in comparing an alternative and a reference procedure, regardless of if a gold standard procedure is available or not. When there is a gold standard, the 2×2 table can represent the correct classification of the two procedures with

respect to the gold standard [18]. In this case, the probabilities p_t and p_r are the accuracy of the test and the reference procedures, respectively. The difference $p_t - p_r$ represents a summary measure of the differences of sensitivity and specificity for these two procedures. In either case, the difference $p_t - p_r$ provides a measure for assessment of an equivalence or non-inferiority between the two procedures.

The equivalence between the alternative and the reference procedures can be tested in terms of the following interval hypotheses:

$$H_0: p_t - p_r \geq \delta \quad \text{or} \quad p_t - p_r \leq -\delta \quad \text{versus} \quad H_1: -\delta < p_t - p_r < \delta \quad (1)$$

where δ is some predetermined clinically meaningful equivalence limit. The equivalence limits do not have to be symmetric about 0 and the results of this paper can be easily generalized to the situation where the lower and upper equivalence limits are not symmetric.

The interval hypotheses can then be decomposed into two sets of one-sided hypotheses:

$$H_{0l}: p_t - p_r \leq -\delta \quad \text{versus} \quad H_{1l}: p_t - p_r > -\delta \quad (2)$$

and

$$H_{0u}: p_t - p_r \geq \delta \quad \text{versus} \quad H_{1u}: p_t - p_r < \delta \quad (3)$$

The one-sided hypothesis (2) has been referred as to the non-inferiority hypothesis.

2.2. Asymptotic tests

Let $\theta = p_t - p_r = p_{10} - p_{01}$ denote the difference of response probabilities between the alternative and the reference procedures. Denote the sample estimates of p_{10} and p_{01} by $\hat{p}_{10} = x_{10}/n$ and $\hat{p}_{01} = x_{01}/n$, respectively. Under the trinomial model, the estimate $\hat{\theta} = \hat{p}_{10} - \hat{p}_{01}$ has mean θ and variance $\sigma^2 = [(p_{01} + p_{10}) - \theta^2]/n$. A simple approach to estimate σ^2 is to replace the parameters by their sample estimates, that is, $\hat{\sigma}^2 = [(\hat{p}_{01} + \hat{p}_{10}) - \hat{\theta}^2]/n$. The $\hat{\sigma}^2$ also is the ML estimate without constraint. A Wald-type asymptotic test for hypothesis (2) at the level α is given by

$$Z_l = \frac{\hat{\theta} + \delta}{\hat{\sigma}} = \frac{c + n\delta}{\sqrt{(d - n\hat{\theta}^2)}} \geq z_\alpha \quad (4)$$

where $c = x_{10} - x_{01}$ and $d = x_{01} + x_{10}$, and z_α is the upper 100 α per cent percentile of the standard normal distribution. Similarly, the test for the hypothesis (3) is given by

$$Z_u = \frac{\hat{\theta} - \delta}{\hat{\sigma}} = \frac{c - n\delta}{\sqrt{(d - n\hat{\theta}^2)}} \leq -z_\alpha \quad (5)$$

Based on the intersection-union principle [19], the two one-sided tests procedure rejects the hypothesis H_0 and declares equivalence of two treatments if both tests Z_l and Z_u are rejected. For the non-inferiority test, as pointed out by one of the referees, the ICH E9 guideline Statistical Principles for Clinical Trials [1] suggests using a significance level of $\alpha/2$ for the hypothesis (2). It follows that the alternative procedure is claimed to be not inferior to the reference procedure if $Z_l \geq z_{\alpha/2}$.

Two other approaches have been proposed to estimate the variance σ^2 in the context of non-inferiority testing. Lu and Bean [16] proposed an approach by substituting θ^2 with δ^2 under the null hypothesis constraint, that is, $\hat{\sigma}^2 = [(\hat{p}_{01} + \hat{p}_{10}) - \delta^2]/n$. Nam [17] pointed out that this approach is anticonservative. Nam [17] further proposed an alternative approach by the restricted maximum likelihood estimation (RMLE), that is, the score method [20]. Under the inferiority hypothesis (2), the RMLE of σ^2 is $\tilde{\sigma}_1^2 = [(\tilde{p}_{1,01} + \tilde{p}_{1,10}) - \delta^2]/n$, where $\tilde{p}_{1,10}$ and $\tilde{p}_{1,01}$ are the RMLEs evaluated at the boundary, $p_{10} - p_{01} = -\delta$, that is, $\tilde{p}_{1,01} = \{-\tilde{a}_1 + (\tilde{a}_1^2 - 8\tilde{b}_1)^{1/2}\}/4$ and $\tilde{p}_{1,10} = \tilde{p}_{1,01} - \delta$, with $\tilde{a}_1 = -\hat{\theta}(1 - \delta) - 2(\hat{p}_{01} + \delta)$ and $\tilde{b}_1 = \delta(1 + \delta)\hat{p}_{01}$. Similarly, under the null hypothesis (3), the restricted MLE of σ^2 at the boundary $p_{10} - p_{01} = \delta$ is $\tilde{\sigma}_u^2 = [(\tilde{p}_{u,01} + \tilde{p}_{u,10}) - \delta^2]/n$, where $\tilde{p}_{u,01} = \{-\tilde{a}_u + (\tilde{a}_u^2 - 8\tilde{b}_u)^{1/2}\}/4$, $\tilde{p}_{u,10} = \tilde{p}_{u,01} + \delta$, and $\tilde{a}_u = -\hat{\theta}(1 + \delta) - 2(\hat{p}_{01} - \delta)$ and $\tilde{b}_u = -\delta(1 - \delta)\hat{p}_{01}$. The RMLE testing procedure can then be obtained by substituting $\hat{\sigma}$ in (4) and (5) by $\tilde{\sigma}_1$ and $\tilde{\sigma}_u$, respectively. In this paper, we focus on the two estimates $\hat{\sigma}^2$ and $(\tilde{\sigma}_1^2, \tilde{\sigma}_u^2)$ to test for equivalence or non-inferiority. We will refer to the former as the sample-based test and to the latter as the RMLE-based test.

As discussed, an equivalence study is often tested by the confidence interval approach. That is, the equivalence between the test and reference procedure is concluded at the α level if and only if the $100(1 - 2\alpha)$ per cent confidence interval is within $(-\delta, \delta)$. The sample-based $100(1 - 2\alpha)$ per cent two-sided confidence interval for the difference between the two response rates is given as $(\hat{\theta} - z_\alpha \hat{\sigma}, \hat{\theta} + z_\alpha \hat{\sigma})$. Alternatively, a $100(1 - 2\alpha)$ per cent asymptotic RMLE-based confidence interval can be obtained, denoted as $(\tilde{\theta}_l, \tilde{\theta}_u)$, where

$$\tilde{\theta}_l = \inf \left\{ \theta_0 : \frac{(\hat{\theta} - \theta_0)}{\tilde{\sigma}_{\theta_0}} < z_\alpha \right\} \quad \text{and} \quad \tilde{\theta}_u = \sup \left\{ \theta_0 : \frac{(\hat{\theta} - \theta_0)}{\tilde{\sigma}_{\theta_0}} > -z_\alpha \right\}$$

with $\tilde{\sigma}_{\theta_0}$ the RMLE of σ at $\theta = \theta_0$. Note that $\tilde{\sigma}_l = \tilde{\sigma}_{-\delta}$ and $\tilde{\sigma}_u = \tilde{\sigma}_\delta$. The RMLE-based confidence interval can be obtained iteratively.

The normal approximation to the binomial distribution is often used with a continuity correction, especially when the sample size is small. The general forms of the two one-sided test statistics for the difference with a continuity correction are

$$Z_l = \frac{c + n\delta - 1/2}{\sqrt{\{n^2 \text{var}(\hat{\theta})\}}} \quad \text{and} \quad Z_u = \frac{c - n\delta + 1/2}{\sqrt{\{n^2 \text{var}(\hat{\theta})\}}}$$

3. TYPE I ERROR AND POWER FUNCTIONS

3.1. True type I error and power

The true type I error rate and power of the asymptotic tests considered above can be computed by enumeration of the exact probabilities from the trinomial distribution

$$P_l(p_{10}, p_{01}) = \frac{n!}{x_{10}!x_{01}!(n - x_{10} - x_{01})!} p_{10}^{x_{10}} p_{01}^{x_{01}} (1 - p_{10} - p_{01})^{n - (x_{01} + x_{10})} \quad (6)$$

For the significance level of $\alpha = 0.05$, the rejection region of the non-inferiority hypothesis H_{01} with critical value $z_{\alpha/2}$ as suggested by ICH E9 is

$$R_l(\alpha = 0.025) = \{(x_{10}, x_{01}) | Z_l(x_{10}, x_{01}) \geq 1.96\}$$

and the rejection region of the equivalence test is

$$R_e(\alpha = 0.05) = \{(x_{10}, x_{01}) | Z_l(x_{10}, x_{01}) \geq 1.64 \text{ and } Z_u(x_{10}, x_{01}) \leq -1.64\}$$

The type I error rate and power can be defined in terms of rejection regions. For example, the probability of rejecting the null hypothesis H_{01} is

$$\sum_{x_{10}=0}^n \sum_{x_{01}=0}^n \frac{n!}{x_{10}! x_{01}! (n - x_{10} - x_{01})!} p_{10}^{x_{10}} p_{01}^{x_{01}} (1 - p_{10} - p_{01})^{n - (x_{10} + x_{01})} I_{[(x_{10}, x_{01}) \in R_1(z)]}$$

We assume that $0 < p_{01} < 1$. When $p_{10} = 0$, the probability of rejection becomes the binomial

$$\sum_{x_{01}=0}^n \frac{n!}{x_{01}! (n - x_{01})!} p_{01}^{x_{01}} (1 - p_{01})^{n - x_{01}} I_{[(0, x_{01}) \in R_1(z)]}$$

Note that, for the sample point $(x_{01}, x_{10}) = (0, 0)$, the sample-based test statistics have values $Z_l = \infty$ and $Z_u = -\infty$. Thus, the point $(0, 0)$ is always included in the rejection region for both of the two one-sided tests. The sample-based test statistics also have values $-\infty$ and ∞ , respectively at $(x_{01}, x_{10}) = (n, 0)$ and $(x_{01}, x_{10}) = (0, n)$. The two points are always excluded from the rejection regions for testing equivalence. However, for testing non-inferiority, the point $(x_{01}, x_{10}) = (0, n)$ should be included in the rejection region.

The true type I error rate and power were calculated for various combinations of the sample sizes ($n = 30, 50, 100, 200$), equivalence limits ($\delta = 0.1, 0.15, 0.2$), and the binary probabilities for the reference procedure ($p_{01} = 0.05, 0.1, 0.15, 0.2, 0.3$). The probabilities for the alternative procedure were $p_{10} = p_{01} - \delta$ for the non-inferiority test and $p_{10} = p_{01} \pm \delta$ for the equivalence test.

Table I contains the actual type I error rates and powers of the the sample-based test and RMLE-based test, with and without the continuity correction, for the one-sided non-inferiority test by using the critical value $z_{0.025} = 1.96$. The value at $p_{10} = p_{01} - \delta$ represents the actual level of the test, and the value at $p_{10} = p_{01}$ represents the power. The RMLE-based test controls type I error better than the sample-based test. The RMLE-based test without the continuity correction performs well except for when the response rate of the alternative procedure is at the boundary point when $p_{10} = 0$. The test can become very conservative when it is used with the continuity correction. On the other hand, the sample-based test shows anticonservatism without the continuity correction. With the continuity correction, the actual nominal levels are all below the nominal level except for $p_{10} = 0$.

Table II contains the type I error rates and powers for the two one-sided equivalence tests with nominal level $\alpha = 0.05$. The value at $p_{10} = (p_{01} - \delta)$ or $(p_{01} + \delta)$ represents the actual level of the test, and the value at $p_{10} = p_{01}$ represents the power. In general, for a fixed value of δ , the power increases as p_{01} decreases or n increases. Both tests show very low power for sample sizes 50 and 100 when $\delta = 0.05$ and 0.10. When the sample size is sufficient, the sample-based test seems to have difficulty maintaining the type I error when $p_{10} = 0$ even with the continuity correction. Again, the RMLE-based test without the continuity correction performs reasonably well, except for $p_{10} = 0$. In summary, based on the results given in Tables I and II, the significant level of the asymptotic tests cannot be adequately controlled when the probability of the discordant pair is smaller than 0.20 or the total sample size is fewer than 50.

Table I. Actual significance levels and powers for the one-sided test using the sample-based and RMLE-based variance estimates with and without continuity corrections (CC) for $n = 30, 50, 100$ at $\alpha = 2.5$ per cent.

p_{01}	p_{10}	δ	Without CC						With CC					
			Sample-based			RMLE-based			Sample-based			RMLE-based		
			30	50	100	30	50	100	30	50	100	30	50	100
0.05	0.00	0.05	0.215*	0.077	0.118	0.000	0.000	0.006	0.000	0.077	0.037	0.000	0.000	0.006
		0.05	0.237†	0.257	0.397	0.050	0.120	0.289	0.178	0.185	0.322	0.016	0.072	0.238
0.10	0.05	0.05	0.054	0.039	0.033	0.013	0.017	0.022	0.038	0.024	0.023	0.004	0.011	0.015
		0.10	0.120	0.135	0.205	0.066	0.097	0.179	0.079	0.100	0.179	0.034	0.084	0.155
	0.00	0.10	0.184	0.112	0.058	0.000	0.005	0.024	0.042	0.034	0.058	0.000	0.005	0.008
		0.10	0.288	0.395	0.627	0.183	0.317	0.570	0.213	0.319	0.567	0.119	0.259	0.528
0.15	0.10	0.05	0.038	0.030	0.027	0.020	0.022	0.023	0.022	0.022	0.022	0.010	0.017	0.018
		0.15	0.095	0.107	0.150	0.062	0.090	0.142	0.061	0.085	0.132	0.040	0.066	0.122
	0.05	0.10	0.057	0.042	0.032	0.019	0.022	0.022	0.031	0.027	0.025	0.008	0.015	0.017
		0.15	0.200	0.264	0.462	0.151	0.242	0.438	0.144	0.219	0.414	0.115	0.199	0.403
	0.00	0.15	0.048	0.046	0.055	0.008	0.014	0.027	0.048	0.046	0.027	0.008	0.014	0.012
		0.15	0.348	0.511	0.781	0.304	0.466	0.772	0.283	0.452	0.756	0.220	0.426	0.744
0.20	0.15	0.05	0.035	0.030	0.026	0.022	0.026	0.024	0.021	0.022	0.021	0.015	0.017	0.020
		0.20	0.077	0.088	0.126	0.059	0.084	0.117	0.053	0.072	0.108	0.045	0.059	0.102
	0.10	0.10	0.037	0.031	0.030	0.022	0.024	0.025	0.022	0.021	0.022	0.014	0.018	0.020
		0.20	0.155	0.204	0.356	0.137	0.202	0.354	0.119	0.178	0.326	0.107	0.158	0.321
	0.05	0.15	0.046	0.036	0.031	0.020	0.021	0.024	0.030	0.028	0.025	0.010	0.015	0.019
		0.20	0.265	0.390	0.667	0.257	0.383	0.653	0.232	0.352	0.634	0.198	0.350	0.634
	0.00	0.20	0.044	0.048	0.047	0.011	0.018	0.025	0.044	0.048	0.025	0.011	0.006	0.013
		0.20	0.410	0.626	0.887	0.408	0.596	0.883	0.387	0.563	0.872	0.350	0.558	0.873
0.30	0.10	0.20	0.028	0.028	0.027	0.026	0.023	0.024	0.023	0.021	0.022	0.016	0.018	0.021
	0.30		0.307	0.460	0.732	0.307	0.460	0.742	0.276	0.411	0.712	0.275	0.427	0.722

* Type I error rate.

† Power.

3.2. Power and sample size

This section considers the asymptotic power and sample size required for the two tests described above. The power functions and sample size calculations given by Nam [17] for the one-sided non-inferiority test will be extended to the two one-sided tests for the equivalence test. Detailed derivations are given in the Appendix. Denote $\bar{\sigma}_l$ and $\bar{\sigma}_u$ as the values of estimated standard errors $\hat{\sigma}_l$ and $\hat{\sigma}_u$ with $\tilde{p}_{l,01}, \tilde{p}_{l,10}, \tilde{p}_{u,01}$, and $\tilde{p}_{u,10}$ replaced by their asymptotic limits $\bar{p}_{l,01}, \bar{p}_{l,10}, \bar{p}_{u,01}$, and $\bar{p}_{u,10}$, respectively. Given $\theta = \theta_0$, when n becomes large, $\bar{\sigma}_l$ and $\bar{\sigma}_u$ are given as

$$\bar{\sigma}_l = \sqrt{\{(2\bar{p}_{l,01} - \delta - \delta^2)/n\}}, \quad \bar{\sigma}_u = \sqrt{\{(2\bar{p}_{u,01} + \delta - \delta^2)/n\}}$$

where $\bar{p}_{l,01} = \{-a_l + (a_l^2 - 8b_l)^{1/2}\}/4$ and $\bar{p}_{u,01} = \{-a_u + (a_u^2 - 8b_u)^{1/2}\}/4$, and $a_l = -\theta_0(1 - \delta) - 2(p_{01} + \delta)$, $a_u = -\theta_0(1 + \delta) - 2(p_{01} - \delta)$, $b_l = p_{01}\delta(1 + \delta)$, $b_u = -p_{01}\delta(1 - \delta)$. Since the sample estimate $\hat{\sigma}$ is independent of the null hypothesis, it follows that for the sample-based test, $\bar{\sigma}_l = \bar{\sigma}_u = \sigma$, the true standard error of $\hat{\theta}$. Denote $w_l = \sigma/\bar{\sigma}_l$ and $w_u = \sigma/\bar{\sigma}_u$. The asymptotic

Table II. Actual significance levels and powers for the two one-sided tests using the sample-based and RMLE-based variance estimates with and without continuity corrections (CC) for $n = 50, 100, 200$ at $\alpha = 5$ per cent.

p_{01}	p_{10}	δ	Without CC						With CC					
			Sample-based			RMLE-based			Sample-based			RMLE-based		
			50	100	200	50	100	200	50	100	200	50	100	200
0.10	0.00	0.10	0.112*	0.117	0.093	0.034	0.058	0.057	0.112	0.058	0.057	0.034	0.024	0.032
	0.10		0.077†	0.444	0.870	0.026	0.417	0.861	0.027	0.382	0.852	0.006	0.328	0.832
	0.20		0.003*	0.034	0.052	0.000	0.034	0.051	0.000	0.021	0.045	0.000	0.020	0.043
0.15	0.00	0.15	0.112	0.099	0.065	0.046	0.055	0.041	0.046	0.055	0.065	0.014	0.027	0.041
	0.15		0.257	0.734	0.974	0.191	0.711	0.973	0.185	0.689	0.970	0.121	0.664	0.969
	0.30		0.013	0.050	0.050	0.011	0.048	0.051	0.006	0.042	0.046	0.002	0.041	0.046
0.20	0.10	0.10	0.003	0.034	0.052	0.000	0.034	0.051	0.000	0.021	0.045	0.000	0.020	0.043
	0.20		0.000	0.021	0.452	0.000	0.021	0.451	0.000	0.006	0.410	0.000	0.006	0.410
	0.30		0.000	0.000	0.042	0.000	0.000	0.042	0.000	0.000	0.036	0.000	0.000	0.036
0.20	0.05	0.15	0.062	0.058	0.054	0.041	0.048	0.049	0.044	0.049	0.047	0.030	0.037	0.043
	0.20		0.083	0.543	0.914	0.066	0.529	0.915	0.046	0.490	0.905	0.021	0.485	0.905
	0.35		0.001	0.044	0.050	0.001	0.044	0.051	0.000	0.036	0.045	0.000	0.036	0.047
0.20	0.00	0.20	0.103	0.080	0.063	0.048	0.047	0.043	0.048	0.047	0.063	0.018	0.026	0.043
	0.20		0.441	0.876	0.995	0.458	0.876	0.996	0.380	0.853	0.995	0.371	0.852	0.995
	0.40		0.030	0.050	0.049	0.038	0.051	0.051	0.016	0.042	0.045	0.023	0.044	0.046
0.30	0.15	0.15	0.013	0.050	0.050	0.011	0.048	0.051	0.006	0.042	0.046	0.002	0.041	0.046
	0.30		0.000	0.238	0.729	0.000	0.250	0.735	0.000	0.194	0.710	0.000	0.197	0.716
	0.45		0.000	0.016	0.049	0.000	0.019	0.051	0.000	0.009	0.044	0.000	0.012	0.046
0.30	0.10	0.20	0.050	0.054	0.051	0.051	0.049	0.051	0.041	0.044	0.045	0.036	0.042	0.045
	0.30		0.149	0.662	0.957	0.189	0.669	0.959	0.080	0.620	0.952	0.111	0.634	0.954
	0.50		0.003	0.047	0.048	0.007	0.049	0.051	0.000	0.040	0.045	0.001	0.045	0.048

* Type I error rate.

† Power.

power function of the α -level two one-sided tests procedure for $|\theta_0| < \delta$ is given as

$$\begin{aligned} \beta(\theta_0) &= P(Z_l \geq z_\alpha, Z_u \leq -z_\alpha | \theta_0) \\ &= \begin{cases} \Phi(c_u) - \Phi(c_l), & \text{if } c_u - c_l > 0 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, and $c_u = -\theta_0/\sigma + \delta/\sigma - z_\alpha/w_u$, $c_l = -\theta_0/\sigma - \delta/\sigma + z_\alpha/w_l$. Note that for the sample-based test $w_l = w_u = 1$. For the RMLE-based test

$$w_l = \frac{(2p_{01} + \theta_0 - \theta_0^2)^{1/2}}{(2\bar{p}_{l,01} - \delta - \delta^2)^{1/2}}, \quad w_u = \frac{(2p_{01} + \theta_0 - \theta_0^2)^{1/2}}{(2\bar{p}_{u,01} + \delta - \delta^2)^{1/2}}$$

Similarly, the asymptotic power for the non-inferiority test is $1 - \Phi(-\theta_0/\sigma - \delta/\sigma + z_\alpha/2/w_l)$, for $\theta_0 > -\delta$.

At $\theta_0 = 0$, the interval (c_l, c_u) is symmetric about zero. The sample size required for power $1 - \beta$ for the two one-sided tests procedure is

$$2p_{01} \left\{ \frac{z_\alpha/\bar{w} + z_{\beta/2}}{\delta} \right\}^2 \quad (7)$$

where $\bar{w} = 1$ and $\bar{w} = (2p_{01})^{1/2}/(2\bar{p}_{l,01} - \delta - \delta^2)^{1/2}$ for the sample-based test and RMLE-based test, respectively. With the continuity correction (CC), the sample size n required for the two one-sided tests can be modified as the root of the following equation:

$$n \left(\delta - \frac{1}{2n} \right)^2 = 2p_{01} (z_\alpha/\bar{w} + z_{\beta/2})^2$$

Similar formulae can be obtained for the non-inferiority test by substituting z_α , $z_{\beta/2}$ with $z_{\alpha/2}$ and z_β , respectively.

Table III provides the theoretical sample sizes required to obtain at least 80 per cent power for the various combinations of p_{01} , p_{10} and δ of the one-sided non-inferiority test given in Table I. The sample size is calculated based on the true difference between the two response rates being 0. In addition, the empirical type I error and power for the theoretical sample size were also computed by a Monte Carlo simulation with 10 000 random samples. Table IV was constructed similar to Table III for the two one-sided tests given in Table II. From the empirical results of Tables III and IV, it can be seen that the RMLE-based test controls the type I error better than the sample-based test. The sample-based test is anticonservative especially when p_{01} or p_{10} is near 0. Both Tables III and IV indicate that the sample sizes do provide the desired power. The difference between the RMLE-based test without the continuity correction and the sample-based test with the continuity correction is small. Table IV shows that to establish an equivalence between two procedures with symmetric equivalence limits of 0.15, a minimal sample size of 120 is needed for $p_{01} \geq 0.1$, and with the equivalence limits of 0.10, a minimal sample size of 180 is needed for $p_{01} \geq 0.15$.

4. NUMERICAL EXAMPLE

We consider the example of a clinical trial to compare the non-invasive MRI alternative procedure and the invasive CTAP reference procedure described in Section 1. Briefly, the biopsy histopathologic examination usually provides a definite diagnosis for malignant lesions. It is an invasive surgical procedure with possible post-operative complications. The CTAP imaging procedure for detecting liver malignancy can provide a reasonable accuracy in diagnosis as compared to biopsy. However, this procedure is an invasive and expensive surgical procedure and may also have post-operative complications. Therefore, ultrasound or magnetic resonance imaging in conjunction with the use of a contrast enhancement agent (MRI) has been suggested as a non-invasive diagnosis imaging alternative to the CTAP.

The diagnostic results of the non-invasive MRI test procedure, invasive CTAP reference procedure, and the gold standard from biopsy histopathology are summarized in Table V.

An objective of this trial is to assess whether the diagnostic accuracy provided by the non-invasive MRI procedure is not inferior to that by the invasive CTAP method with an acceptable difference (equivalence limit), as compared to gold standard of the histopathologic

Table III. Sample size, n , and empirical type I error and power, $1 - \hat{\beta}$, for the one-sided test using the sample-based and RMLE-based variance estimates with and without continuity corrections (CC).

p_{01}	p_{10}	δ	Without CC				With CC			
			Sample-based		RMLE-based		Sample-based		RMLE-based	
			n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$
0.05	0.00	0.05	314*	0.0454 [†]	350	0.0186	334	0.0491	370	0.0114
	0.05			0.8091 [‡]		0.8061		0.7938		0.8092
0.10	0.05	0.05	628	0.0258	644	0.0225	648	0.0239	664	0.0221
	0.10			0.8007		0.8073		0.7968		0.8059
	0.00	0.10	157	0.0401	171	0.0190	167	0.0476	180	0.0125
0.15	0.10			0.8018		0.8198		0.7949		0.8067
	0.10	0.05	942	0.0258	949	0.0268	962	0.0218	970	0.0206
				0.7952		0.7932		0.7974		0.8002
	0.05	0.10	236	0.0273	242	0.0243	246	0.0253	252	0.0194
	0.15			0.8027		0.8067		0.7970		0.7983
	0.00	0.15	105	0.0370	111	0.0223	112	0.0375	117	0.0161
0.20	0.15			0.8058		0.8121		0.8081		0.8070
	0.15	0.05	1256	0.0239	1259	0.0246	1276	0.0256	1279	0.0233
				0.7986		0.8061		0.7960		0.7942
	0.20	0.10	314	0.0251	317	0.0270	324	0.0244	327	0.0201
	0.20			0.7992		0.8076		0.7944		0.8066
	0.05	0.150	140	0.0311	142	0.0226	147	0.0243	149	0.0213
	0.20			0.7973		0.8014		0.8013		0.8109
	0.00	0.20	79	0.0579	81	0.0265	84	0.0385	86	0.0120
0.30	0.20			0.7976		0.8059		0.8124		0.8168
	0.10	0.20	118	0.0229	116	0.0255	123	0.0232	121	0.0223
	0.30			0.8027		0.8014		0.7926		0.8040

* Sample size is calculated based on the formula given in Section 3.2 when $\theta_0=0$ for a required power of 80 per cent at the 2.5 per cent significance level.

[†] Empirical type I error.

[‡] Empirical power.

result from biopsy. The historical data suggest that CTAP yields a correct diagnosis rate between 0.80 and 0.90 as compared to histopathological diagnosis. For illustration purposes, two symmetric equivalence limits $\delta=0.10,0.15$, are considered in the analysis. The continuity correction is applied to the sample-based test based on the results of the previous section.

Because the gold standard is available for this trial, the frequency of the correct diagnosis, x_{ij} , $i, j = 1, 0$ (Yes = 1, No = 0), can be obtained as $x_{11} = 37$, $x_{10} = 5$, $x_{01} = 4$ and $x_{00} = 4$. Table VI contains a summary of the analyses. For $\delta=0.15$, the p -values of both one-sided test procedures are all highly significant and the lower bounds of the 97.5 per cent confidence interval are greater than -0.15 . However, for $\delta=0.10$, the p -values are not less than the significance level 0.025. Thus we can conclude that the accuracy measured by the MRI procedure is no more than 0.15 below that measured by the CTAP procedure.

For illustration purpose, if the gold standard is not available, the x_{ij} is the frequency of the diagnostic responses (regardless of the outcomes of the gold standard). The observed

Table IV. Sample size, n , and empirical power, $1 - \hat{\beta}$, for the two one-sided tests using the sample-based and RMLE-based variance estimates with and without continuity corrections (CC).

p_{01}	p_{10}	δ	Without CC				With CC			
			Sample-based		RMLE-based		Sample-based		RMLE-based	
			n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$	n	$1 - \hat{\beta}$
0.10	0.00	0.10	172*	0.0693 [†]	183	0.0410	182	0.0735	193	0.0245
	0.10			0.8013 [‡]						
	0.20			0.0525 [†]						
0.15	0.00	0.15	115	0.0599	120	0.0394	121	0.0682	126	0.0257
	0.15			0.8114						
	0.30			0.0503						
0.20	0.10	0.10	343	0.0504	345	0.0485	353	0.0450	355	0.0463
	0.20			0.8041						
	0.30			0.0539						
0.20	0.05	0.15	153	0.0603	155	0.0507	159	0.0509	161	0.0396
	0.20			0.8031						
	0.35			0.0479						
0.20	0.00	0.20	86	0.0543	88	0.0478	91	0.0612	93	0.0254
	0.20			0.8167						
	0.40			0.0493						
0.30	0.15	0.15	229	0.0515	227	0.0514	235	0.0431	234	0.0471
	0.30			0.8080						
	0.45			0.0489						
0.30	0.10	0.20	129	0.0538	127	0.0508	134	0.0455	132	0.0437
	0.30			0.8101						
	0.50			0.0517						

* Sample size is calculated based on the formula given in Section 3.2 when $\theta_0=0$ for a required power of 80 per cent with a symmetric limits of δ at the 2.5 per cent significance level.
[†] Empirical type I error.
[‡] Empirical power.

Table V. Diagnostic results of the non-invasive MRI test procedure, invasive CTAP reference procedure and the biopsy histopathology.

Gold standard (k)	Test procedure (i)	Reference procedure (j)	Frequency
Yes (1)	Yes (1)	Yes (1)	36
		No (0)	3
	No (0)	Yes (1)	2
		No (0)	1
No (0)	Yes (1)	Yes (1)	3
		No (0)	2
	No (0)	Yes (1)	2
		No (0)	1

responses for the four outcomes become $x_{11} = 39$ $x_{10} = 5$ $x_{01} = 4$ and $x_{00} = 2$. In the absence of a gold standard, an objective of the study is to assess if the non-invasive MRI procedure can be as a reliable diagnostic method as the invasive CTAP procedure within an acceptable

Table VI. Z-scores and p -values of the sample-based test and RMLE-based test and the confidence limits for non-inferiority between post-contrast MRI alternative and CTAP reference procedures.

δ	Test	$\hat{\sigma}/\tilde{\sigma}_l$	Z_l	Tail probability*	97.5 per cent one-sided CI
0.15	Sample	0.0599	2.670	0.0038	(-0.1074, 1)
	RMLE	0.0689	2.469	0.0068	(-0.1188, 1)
0.10	Sample	0.0599	1.835	0.0333	(-0.1074, 1)
	RMLE	0.0654	1.836	0.0332	(-0.1188, 1)

* Tail probability = $1 - \Phi(Z_l)$.Table VII. Z-scores and p -values of the sample-based test and RMLE-based test and the confidence limits for the equivalence between post-contrast MRI alternative and CTAP reference procedures.

δ	Test	$\hat{\sigma}/\tilde{\sigma}_l$	Z_l	$\hat{\sigma}/\tilde{\sigma}_u$	Z_u	95 per cent CI	p -value
0.15	Sample	0.0599	2.670 (0.0038)*	0.0599	-2.002 (0.0226) [†]	(-0.0886, 0.1286)	0.0226
	RMLE	0.0689	2.469 (0.0068)	0.0668	-1.945 (0.0259)	(-0.0859, 0.1274)	0.0259
0.10	Sample	0.0599	1.835 (0.0333)	0.0599	-1.168 (0.1214)	(-0.0886, 0.1286)	0.1214
	RMLE	0.0654	1.836 (0.0332)	0.0635	-1.261 (0.1037)	(-0.0859, 0.1274)	0.1037

* Tail probability = $1 - \Phi(Z_l)$.[†] Tail probability = $\Phi(Z_u)$.

range of equivalent limits. Table VII contains a summary of the analyses. The p -value for the two one-sided tests is the larger of tailed probabilities of the two one-sided tests Z_l and Z_u .

For $\delta=0.15$, since the p -value of the two one-sided tests is less than the 5 per cent nominal level, we can reject the null hypothesis and conclude that the diagnostic efficacy of the post-contrast MRI procedure is equivalent to that of CTAP procedure at the 5 per cent nominal level. For $\delta=0.10$, the p -value is greater than the nominal level, and hence we cannot conclude the two procedures are equivalent.

5. DISCUSSION

Equivalence trials are important in the area of diagnostic imaging. Ideally, diagnostic procedures with high accuracy are most desirable but they are frequently too expensive or hazardous to be used. An invasive procedure and/or a requirement for sophisticated machinery, complicated operating procedures, bug-free computer software, and well-trained radiologists for analysing imaging pictures may be less desirable than a non-invasive diagnostic imaging procedure, which does not require expensive machinery but also produces an equivalent diagnostic efficacy with a better safety profile. In this paper we present two asymptotic tests for assessing equivalence between two diagnostic procedures. The test is a generalization of McNemer's test with a non-zero difference in both upper and lower bounds. As stated by Nam [17] for testing non-inferiority, unlike the McNemer's test, the test statistics take into consideration not only the number of patients with discordant results but also the total number of patients in the study.

Lu and Bean (LB) [16] considered the non-inferiority testing in the matched-pair data for the first time. Nam [17] investigated the performance of the LB test by using the Monte Carlo simulation for testing non-inferiority. He found that the LB test is anticonservative in the manner that its empirical significant level exceeds the nominal level in several cases. Under the alternative hypothesis, the variance estimate $\hat{\sigma}^2$ in the LB test statistic underestimates the true variance σ^2 . The $\hat{\sigma}^2$ employed in the sample-based test statistic is an unbiased estimator of σ^2 . Therefore, $\hat{\sigma}^2$ is likely to be smaller than σ^2 . For the example data, $\hat{\sigma}^2 = (0.0583)^2$ and $(0.0561)^2$ for $\delta = 0.10$ and 0.15 , respectively, while $\sigma^2 = (0.0599)^2$ (see Table VI).

Equivalence limits can be based on one of the three measures: difference, rate ratio, or odds ratio between the two binomial probabilities. Although a criterion based on difference in proportion has been proposed as a standard criterion [8, 21], rate ratio or odds ratio are being considered as potential criteria for evaluation of equivalence or non-inferiority. In some applications, the rate ratio or odds ratio measure may be more applicable when the response rates are close to 0 or 1. The interval hypothesis in terms of the risk ratio can be expressed as

$$H_0: p_t/p_r \geq c \quad \text{or} \quad p_t/p_r \leq 1/c \quad \text{versus} \quad H_1: 1/c < p_t/p_r < c$$

where $p_r > 0$, $c > 1$. The interval hypothesis can be restated as the two one-sided hypotheses:

$$H_{0l}: p_t - p_r/c \leq 0 \quad \text{versus} \quad H_{1l}: p_t - p_r/c > 0$$

and

$$H_{0u}: p_t - p_r c \geq 0 \quad \text{versus} \quad H_{1u}: p_t - p_r c < 0$$

The two one-sided tests procedure for the ratio of two binomial probabilities can similarly be derived in terms of the difference of the two probabilities described in this paper. A Wald-type (sample-based) statistic for testing equivalence based on the rate ratio was proposed by Lachenbruch and Lynch [18]. Comparisons of the two tests will be of interest for future investigations.

It has been argued that asymptotic tests are unreliable when the sample size is small. Table I shows that the RMLE-based test has type I error rates not exceeding the 0.05 nominal level except when $n = 100$ and $p_{01} = 0.10, 0.15$ at $p_{10} = 0$; $p_{01} = 0.20$, $p_{10} = 0.10$; and $p_{01} = 0.30$, $p_{10} = 0.10$. The test with the continuity correction becomes conservative. On the other hand, when $n = 30$ or 50 , the test without the continuity correction seems to control the type I error reasonably well (with very few exceptions). For a small sample size the unconditional exact test may be an alternative approach and requires further research. Finally, Tables III and IV show that when the sample size is sufficient, the type I error rates from the RMLE-based test without the continuity correction are close to or below the nominal level. The RMLE-based test without the continuity correction appears to be an acceptable approach for establishing the equivalence or non-inferiority between two paired binary endpoints.

APPENDIX: DERIVATIONS OF THE ASYMPTOTIC POWER FUNCTION AND SAMPLE SIZE

At $\theta = \theta_0$ and $|\theta_0| < \delta$, let $w_1 = \sigma/\bar{\sigma}_1$, $w_u = \sigma/\bar{\sigma}_u$, the two one-sided tests statistics are approximately

$$Z_1 \approx w_1 \frac{\hat{\theta} + \delta}{\sigma} = w_1 \left(\frac{\hat{\theta} - \theta_0}{\sigma} + \frac{\theta_0 + \delta}{\sigma} \right) \approx w_1 \left(Z + \frac{\theta_0 + \delta}{\sigma} \right)$$

and

$$Z_u \approx w_u \frac{\hat{\theta} - \delta}{\sigma} = w_u \left(\frac{\hat{\theta} - \theta_0}{\sigma} + \frac{\theta_0 - \delta}{\sigma} \right) \approx w_u \left(Z + \frac{\theta_0 - \delta}{\sigma} \right)$$

where Z is a standard normal distributed random variable. As $n \rightarrow \infty$, the asymptotic power function is

$$P(Z_1 \geq z_\alpha, Z_u \leq -z_\alpha | \theta_0) \approx \Phi \left(-\frac{\theta_0}{\sigma} + \frac{\delta}{\sigma} - \frac{z_\alpha}{w_u} \right) - \Phi \left(-\frac{\theta_0}{\sigma} - \frac{\delta}{\sigma} + \frac{z_\alpha}{w_1} \right),$$

provided that $2\delta/\sigma - z_\alpha/w_1 - z_\alpha/w_u > 0$.

At $\theta_0 = 0$, the power function can be expressed as

$$\Phi \left(\frac{\delta}{\sigma} - \frac{z_\alpha}{w_u} \right) - \Phi \left(-\frac{\delta}{\sigma} + \frac{z_\alpha}{w_1} \right),$$

where $\sigma = (2p_{01}/n)^{1/2}$. For the RMLE-based test, the limits $\bar{\sigma}_1$ and $\bar{\sigma}_u$ in w_1 and w_u , respectively, are $\bar{\sigma}_1 = \sqrt{\{(2\bar{p}_{1,01} - \delta - \delta^2)/n\}}$ and $\bar{\sigma}_u = \sqrt{\{(2\bar{p}_{u,01} + \delta - \delta^2)/n\}}$. The limits $\bar{p}_{1,01}$, $\bar{p}_{u,01}$ can be simplified to

$$\bar{p}_{1,01} = \frac{p_{01} + \delta + (p_{01}^2 + \delta^2 - 2p_{01}\delta^2)^{1/2}}{2} \quad \text{and} \quad \bar{p}_{u,01} = \frac{p_{01} - \delta + (p_{01}^2 + \delta^2 - 2p_{01}\delta^2)^{1/2}}{2}$$

It follows that

$$\bar{\sigma}_1 = \bar{\sigma}_u = [\{p_{01} - \delta^2 + (p_{01}^2 + \delta^2 - 2p_{01}\delta^2)^{1/2}\}/n]^{1/2}$$

and $w_1 = w_u \equiv \bar{w}$. The interval $(-\delta/\sigma + z_\alpha/\bar{w}, \delta/\sigma - z_\alpha/\bar{w})$ is symmetric about zero. Then, the sample size required to attain the power $1 - \beta$ at $\theta_0 = 0$ is given as

$$n \geq 2p_{01} \left(\frac{z_\alpha/\bar{w} + z_{\beta/2}}{\delta} \right)^2$$

where $\bar{w} = 1$ and

$$\bar{w} = \frac{(2p_{01})^{1/2}}{(2\bar{p}_{1,01} - \delta - \delta^2)^{1/2}}$$

for the sample-based test and for RMLE-based test, respectively.

REFERENCES

1. Lewis JA. Statistical principles for clinical trials (ICH E9) an introductory note on an international guideline. *Statistics in Medicine* 1999; **18**:1903–1942.
2. Chow SC, Liu JP. *Design and Analysis of Bioavailability and Bioequivalence Studies*. Marcel Dekker: New York, 1992.
3. Röhm J. Therapeutic equivalence investigations: statistical consideration. *Statistics in Medicine* 1998; **17**: 1703–1714.
4. Hilden J, Glasziou P. Regret graphs, diagnostic uncertainty and Youden's index. *Statistics in Medicine* 1996; **15**:969–986.
5. Biggerstaff, BJ. Comparing diagnostic tests: a simple graphic using likelihood ratios. *Statistics in Medicine* 2000; **19**:649–663.
6. Dunnett CW, Gent M. Significance testing to establish equivalence between treatments with special reference to data in the form of 2×2 tables. *Biometrics* 1977; **33**:593–602.
7. Blackwelder WC. Proving the null hypothesis in clinical trials. *Controlled Clinical Trials* 1982; **3**:345–353.
8. Farrington CP, Manning G. Test statistics and sample size formulae for comparative binomial trials with null hypothesis of non-zero risk difference or non-unity relative risk. *Statistics in Medicine* 1990; **9**:1447–1454.
9. Tu D. A comparative study of some statistical procedures in establishing therapeutic equivalence of nonsystemic drugs with binary endpoints. *Drug Information Journal* 1997; **31**:1291–1300.
10. Chen JJ, Tsong Y, Kang S-h. Tests for equivalence and non-inferiority between two proportions. *Drug Information Journal* 2000; **34**:569–578.
11. McNemar S. Note on the sampling error of the differences between corrected proportions and percentages. *Psychometrika* 1947; **12**:153–157.
12. Miettinen OS. The matched-pairs design in the case of all-or-none response. *Biometrics* 1968; **24**:339–352.
13. Connor RJ. Sample size for testing differences in proportions for the paired-sample design. *Biometrics* 1987; **43**:207–211.
14. Suissa S, Shuster JJ. The matched-pairs trial: exact unconditional design and analysis. *Biometrics* 1991; **47**: 361–372.
15. Agresti A. *Categorical Data Analysis*. Wiley: New York, 1990.
16. Lu Y, Bean JA. On the sample size for one-sided equivalence of sensitivities based upon McNemar's test. *Statistics in Medicine* 1995; **14**:1831–1839.
17. Nam J-m. Establishing equivalence of two treatments and sample size requirements in matched-pairs design. *Biometrics* 1997; **53**:1422–1430.
18. Lachenbruch PA, Lynch CJ. Assessing screening tests: extensions of McNemar's test. *Statistics in Medicine* 1998; **17**:2207–2217.
19. Berger RL, Hsu JC. Bioequivalence trials, intersection-union tests and equivalence confidence sets. *Statistical Science* 1996; **11**:283–318.
20. Tango T. Equivalence test and confidence interval for the difference in proportions for the paired-sample design. *Statistics in Medicine* 1998; **17**:891–908.
21. FDA. Guidance – Topical Dermatologic Corticosteroids: In vivo bioequivalence. Office of Generic Drugs, Center for Drug Evaluation and Research, Food and Drug Administration, Rockville, Maryland, 1995.